# Structural Change in the Global Economy*

Hayato Kato[†], Kensuke Suzuki[‡], and Motoaki Takahashi[§]

June 23, 2024

**Abstract**

In the last few decades, advanced countries have witnessed a significant decline in their manufacturing sectors along with emerging economies being integrated into the global economy. Even had such globalization not been for, the fall in manufacturing might have been inevitable because of structural change, a stylized fact of economic growth that economies shift from agriculture to manufacturing and then to service. What role does structural change play in determining the impact of globalization? To address the question, we developed a quantitative dynamic general equilibrium model of trade with capital accumulation *á la* Eaton et al. (2016) and Caliendo and Parro (2015). Our model features nonhomothetic CES preferences developed by Comin et al. (2021), allowing consumers to present varying income elasticities of demand across sectors. We bring the model to the data for the world economy, encompassing three sectors (agriculture, manufacturing, and service) and 24 countries. We calibrate the model's fundamentals, including trade costs and productivity, and solve the model for the transition path. By applying counterfactual trade costs for different sectors, productivity levels, and preferences to the model, we discuss how these factors collectively shape structural change and its interaction with international trade in advanced countries. Specifically, our quantitative model suggests that the decline in the value-added share of manufacturing in the US since 2000 is fully attributable to the China shock.

*JEL Classification*: F1 (Trade), O1 (Economic Development)

*We would like to thank Jonathan Eaton, Philip Sauré, Klaus Wälde and seminar participants at Clark University, JGU Mainz, Bielefeld University, University of Freiburg, Nagoya University, Göttinger Workshop, Midwest Economic Association Annual Conference, Midwest International Trade Conference, CITP Conference, Kansai Macro Workshop, Yonsei University for their helpful comments. This work is supported by the Project Research Grant (IERPK2321), the Institute of Economic Research, Hitotsubashi University; the Japan Society for the Promotion of Science (JP20H01495); and Clark University Start-Up Research Fund.

[†]Graduate School of Economics, Osaka University; CESifo Research Network. Email: hayato.kato.econ@osaka-u.ac.jp

[‡]Department of Economics, Clark University; Economic Research Center, Nagoya University. Email: ksuzuki@clarku.edu

[§]Johannes Gutenberg University Mainz; Kiel Institute for the World Economy. Email: mtakahas@uni-mainz.de

# 1 Introduction

One of the most heated discussions in the last few decades in academic and policy arenas is the significant impacts of globalization on developed countries. The 1990s witnessed the United States (US) lowering trade barriers against Mexico under the North American Free Trade Agreement (NAFTA). In the 2000s, particularly noteworthy are China's accession to the World Trade Organization (WTO) in 2001 and the eastward enlargement of the European Union (EU). A number of studies find adverse effects of these events on industries in developed economies, in particular, their manufacturing employment. Looking at the impacts of China's growing trade, for example, Acemoglu et al. (2016) report 2.0 to 2.4 million US manufacturing workers losing their jobs due to Chinese import competition over 1999 to 2011.[1] Figure 1 (a) shows the evolution of manufacturing value-added share in GDP of selected countries over the past five decades, from 1965 to 2014. The sharp drop in US manufacturing is evident after the 2000s, and one may argue this can be largely attributed to the "China shock."

From the viewpoint of long-term economic development, however, the manufacturing sector in developed countries is bound to shrink even if the impacts of international trade are not taken into account. As the nation's income grows, it reallocates resources from agriculture to manufacturing, and then to service in a process known as structural change (Kuznets, 1973; Herrendorf et al., 2014). Figure 1 illustrates this point; manufacturing value-added share in the US and Germany started to decline in the 1960s (Figure 1 (a)), whereas service value-added share steadily increased over time (Figure 1 (b)). Japan and China exhibit a similar pattern with a time lag.

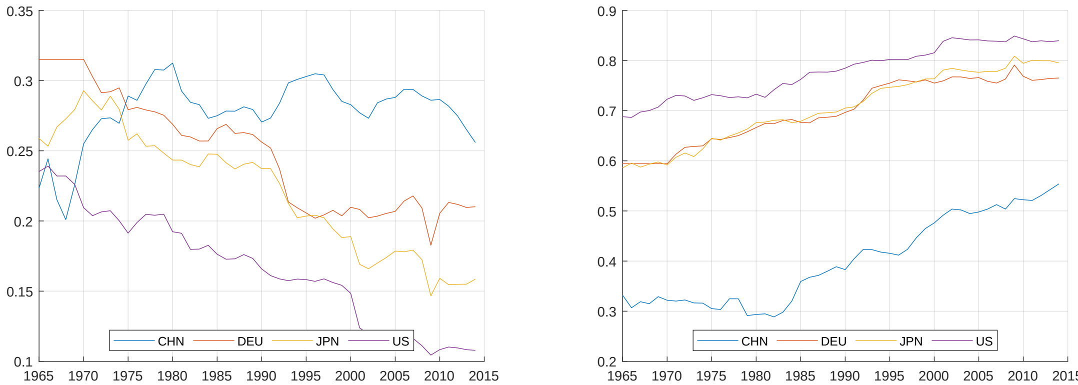
These different views on declining manufacturing in developed countries lead to a number of important questions: What is the role of structural change in determining the impact of growing international trade on resource allocation between sectors? Specifically, to what extent can changes in manufacturing value-added share be attributed to trade and structural change? Whether structural change strengthens or weakens the trade impact in different countries, and if so, how much? How does structural change affect the trade impact on the redistribution of labor and capital income in each sector?

To answer these questions, we propose a unified framework of trade, structural change, and endogenous capital accumulation. We model trade based on the Ricardian comparative advantage *à la* Eaton and Kortum (2002); Caliendo and Parro (2015) and also highlight two prominent drivers of structural change pointed out in the literature: the Baumol effect and the income effect (see for Acemoglu, 2008, Ch.20 a survey). The Baumol effect refers

[1] See also Autor et al. (2013); Acemoglu et al. (2016); Pierce and Schott (2016). The adverse effects of rising Chinese imports are observed in other developed countries, including Canada (Albouy et al., 2019), Denmark (Utar, 2018), France (Malgouyres, 2017), and Germany (Dauth et al., 2014). See also Autor et al. (2016) for a comprehensive review. We note, however, that not all studies find negative effects of rising Chinese imports on developed countries. Taniguchi (2019), for example, finds a positive effect of the China shock in Japan due to the complementary role of imported intermediate inputs.

Figure 1: Value Added Share in GDP

(a) Manufacturing (b) Service




*Notes*: The data is from the WIOD database. See Section 4 for details.

to the substitution of production due to changes in relative prices associated with sector-biased technological change (Baumol, 1967; Ngai and Pissarides, 2007). We capture this by allowing sector-biased technological change and changes in the composition of sectoral inputs in production and investment as in García-Santana et al. (2021); Herrendorf et al. (2021). On the other hand, the income effect emphasizes the non-homothetic preference such that, as income grows, consumers shift their expenditure from the less-income-elastic goods such as foods to the more-income-elastic goods such as services (Kongsamut et al., 2001). Our model employs the nonhomothetic CES preference as in Hanoch (1975); Matsuyama (2019); Comin et al. (2021).[2] Finally, we model forward-looking decisions on capital investment as in Eaton et al. (2016); Ravikumar et al. (2019).

We calibrate our model using the data for 24 countries over half the century, from 1965 to 2014. Our quantification strategy calibrates the model's fundamentals, such as sectoral productivity and trade costs, which allows us to solve the transition paths of the economy in terms of *level*, not in *relative change* known as hat-algebra method (Dekle et al., 2008; Caliendo and Parro, 2015; Caliendo et al., 2019). We then conduct a counterfactual analysis to examine the role of international trade in shaping the industrial structure of advanced economies.

Specifically, among other counterfactual experiments, we ask what the sectoral composition of the economy would look like if the productivity of China and the trade costs between China and the other countries did not change since 1999. This counterfactual experiment teases out the effect of the China shock on the industry composition across developed countries. The

[2]Unlike more standard classes of preference such as the Stone-Geary, the nonhomothetic CES preference shows non-vanishing non-unitary income elasticities as income grows, which is consistent with the finding of Comin et al. (2021), while maintaining analytical tractability as much as possible.

China shock has heterogeneous impacts on the valued-added shares of manufacturing across countries. In Germany and the United States, for example, the counterfactual (no-China-shock) manufacturing shares are higher than the baseline ones by about 2% points and 7% points, respectively, in 2005. In contrast, in Japan, the counterfactual manufacturing share is actually lower than the baseline one by about 2% points in the year. Put differently, the China shock reduced manufacturing in the US and Germany, while it retained manufacturing in Japan. Our counterfactual predictions imply that international trade with China played a big role in accelerating the shift from manufacturing to service in some advanced countries like the US and Germany. That is, traditional structural change forces operated in closed economies, i.e., the income effect and the Bamoul effect, alone cannot explain the rapid deindustrialization at least in the US and Germany. Our predictions also align with results from empirical studies such as Autor et al. (2013) and Taniguchi (2019).

Our study is positioned in the recent literature on quantitative models of structural change embedding international trade (see Alessandria et al., 2023 for a survey). Those studies show a number of new insights such as the decomposition of different mechanisms for declining manufacturing share (Święcki, 2017; Smitkova, 2023), a systematic relationship between countries' intermediate-input intensities and their level of development (Sposi, 2019), and the negative effect of structural change on trade (Lewis et al., 2022). In modeling international trade, these studies follow static models of Eaton and Kortum (2002) and Caliendo and Parro (2015).

Unlike those studies, we propose a dynamic model with endogenous capital accumulation *à la* Eaton et al. (2016) and Ravikumar et al. (2019), which allows us to explain the evolution of sectoral production and production as endogenous outcomes rather than calibration results. This is far from a trivial extension since recent studies in the closed-economy context emphasize the role of investment (e.g., Herrendorf et al., 2014, Sec. 6.3.3) in shaping the industrial structure of the economy. Moreover, our model can speak to the dynamics of trade balance and how this interacts with structural change (Kehoe et al., 2018).

Our paper is most closely related to Sposi et al. (2021), which develops and applies the dynamic model of international trade to study structural change.[3] Despite the similarity in the analytical framework, we have a different set of questions. Sposi et al. (2021) focus on the recent empirical finding of "premature deindustrialization," which conceptualizes that developing countries today experience the transition from manufacturing to service much earlier than those several decades ago (Rodrik, 2016). Furthermore, they also show new

[3]Another closely related study is Święcki (2017) examining to what extent each elements of the model contributes to changes in sectoral composition. He finds that the most important element is the sector-biased productivity. We depart from his static model with non-tradable services by allowing for endogenous capital accumulation and tradable services. His finding might be due to his calibration based on the model without capital, because the estimates of sector-biased sectoral productivity potentially include the contribution by capital. We instead model capital explicitly and give more precise estimates of productivities.

evidence that the cross-country dispersion of manufacturing share has increased over time. Our paper contrasts with their study by emphasizing more country- and region-specific episodes of globalization. While the current manuscript only presents a counterfactual exercise with changing the trade costs universally, we plan to focus on the specific trade shock episolodes, such as WTO accession of China and the EU enlargement, and their impacts on the evolution of the industrial structure of the advanced economies. Furthermore, we to evaluate the welfare effects of these episodes by measuring the equivalent variation of trade shocks and provide a novel decomposition into the terms of trade effect and other effects in an analytical form. We then attempt to quantitatively evaluate each effect and investigate if there are systematic relationships between welfare effects and shifts in sectoral expenditure/production.

The remainder of this paper is structured as follows: section 2 presents the model, section 3 discusses qualitative results of the model, section 4 introduces the calibration of the model and solution algorithm, section 5 presents the quantitative results, and section 6 concludes.

## 2 Model

We consider a dynamic economy in which time is descrete $t = 0, 1, \cdots$. The set of countries is $\mathcal{N} = \{1, 2, \cdots, N\}$. Thus the cardinality of $\mathcal{N}$ is $N$. Countries are generically indexed by $i$ or $n$. There are three sectors: agriculture, manufacturing, and services. Sectors are generically intexed by $j = a, m, s$, where $a$, $m$, and $s$ stand for agriculture, manufacturing, and services, respectively. Sectors and industries are synonymous in this paper.

The representative household in country $n$ as of period 0 maximizes the lifetime utility function

$$\sum_{t=0}^{\infty} \beta^t \zeta_{n,t} L_{n,t} \frac{(C_{n,t}/L_{n,t})^{1-\psi}}{1-\psi} \tag{1}$$

where $\beta \in (0, 1)$ is the discount factor, $\theta > 1$ is the intertemporal elasticity of substitution, $\zeta_{n,t}$ is the demand shifter in country $n$ and period $t$, and $L_{n,t}$ is the population of country $n$ and period $t$. The aggregate consumption in country $n$ and period $t$, $C_{n,t}$, is *implicitly* defined by

$$\sum_{j=a,m,s} (\Omega^j)^{\frac{1}{\sigma}} \left( \frac{C_{n,t}}{L_{n,t}} \right)^{\frac{\epsilon^j (1-\sigma)}{\sigma}} \left( \frac{C^j_{n,t}}{L_{n,t}} \right)^{\frac{\sigma-1}{\sigma}} = 1, \tag{2}$$

where, for $j = a, m, s$, $C^j_{n,t}$ is the composite good of sector $j$ which the representative household in country $n$ and period $t$ consumes, $\Omega^j$ is the demand shifter for sector $j$, $\epsilon^j$ is the parameter governing how the period utility changes as the composite good of sector $j$ changes (nonhomotheticity), and $\sigma$ is the (intratemporal) elasticity of substituion across the sectoral composite goods. The period utility function (2) follows Hanoch (1975) and Comin et al. (2021). Following Eaton and Kortum (2002), we assume that a unit continuum of varieties exists in each sector. For $j = a, m, s$, the composite good of sector $j$ which country $n$ consumes in period $t$ is defined to be

$$C^j_{n,t} = \left[ \int_0^1 C^j_{n,t}(z)^{\frac{\eta-1}{\eta}} dz \right]^{\frac{\eta}{\eta-1}},$$

where $\eta$ is the elasticity of substitution across varieties within sectors.

Solving the intratemporal expenditure minimization problem given $C_{n,t}$, the expenditure of country $n$ in period $t$ is

$$E_{n,t} = L_{n,t} \left[ \sum_{j=a,m,s} \Omega^j_{n,t} \left\{ \left( \frac{C_{n,t}}{L_{n,t}} \right)^{\epsilon^j} P^j_{n,t} \right\}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}, \tag{3}$$

where $P^j_{n,t}$ is the price of the composite good of sector $j$ in country $n$ and period $t$. Define $P_{n,t}$ by $P_{n,t} = E_{n,t}/C_{n,t}$. Then we have

$$P_{n,t} = \left[ \sum_{j=a,m,s} \Omega^j_{n,t} (P^j_{n,t})^{1-\sigma} \left( \frac{C_{n,t}}{L_{n,t}} \right)^{(1-\sigma)(\epsilon^j-1)} \right]^{\frac{1}{1-\sigma}}.$$

The consumption of the composite good of sector $j$ is

$$C^j_{n,t} = L_{n,t} \Omega^j_{n,t} \left( \frac{P^j_{n,t}}{P_{n,t}} \right)^{-\sigma} \left( \frac{C_{n,t}}{L_{n,t}} \right)^{(1-\sigma)\epsilon^j + \sigma}.$$

Let $\omega^j_{n,t}$ be country $n$'s expenditure share on sector $j$ in period $t$, that is, $\omega^j_{n,t} = E^j_{n,t}/E_{n,t}$, where $E^j_{n,t}$ denotes country $n$'s expenditure on sector $j$ goods (or services) in period $t$. Then we have

$$\omega^j_{n,t} = \frac{\Omega^j_{n,t} \left\{ \left( \frac{C_{n,t}}{L_{n,t}} \right)^{\epsilon^j} P^j_{n,t} \right\}^{1-\sigma}}{\sum_{j'=a,m,s} \Omega^{j'}_{n,t} \left\{ \left( \frac{C_{n,t}}{L_{n,t}} \right)^{\epsilon^{j'}} P^{j'}_{n,t} \right\}^{1-\sigma}}, \tag{4}$$

and

$$\frac{\omega^{j'}_{n,t}}{\omega^j_{n,t}} = \left( \frac{P^{j'}_{n,t}}{P^j_{n,t}} \right)^{1-\sigma} \left( \frac{C_{n,t}}{L_{n,t}} \right)^{(1-\sigma)(\epsilon_{j'}-\epsilon_j)} \left( \frac{\Omega^{j'}}{\Omega^j} \right).$$

By definition, $\sum_{j=a,m,s} \omega^j_{n,t} = 1$. See Comin et al. (2021) for detailed derivations of (3) and (4).

The representative household in country $n$ is the sole owner of labor and capital there.

The budget constraint of country $n$ in period $t$ is

$$E_{n,t} + P^K_{n,t} I_{n,t} \leq (1-\phi_{n,t}) \left( w_{n,t} L_{n,t} + r_{n,t} K_{n,t} + \widetilde{T}_{n,t} \right) + L_{n,t} T^P_t, \tag{5}$$

where $P^K_{n,t}$ is the capital good price index which will be defined later, $I_{n,t}$ is the quantity of investment, $\phi_{n,t}$ is the fraction of the aggregate income accrued to the global portfolio, and $T^P_t$ is the payment from the global portfolio to each person of country $n$ in period $t$, and $\widetilde{T}_{n,t}$ is the tariff revenues in country $n$ and period $t$. $\phi_{n,t}$ is an exogenous parameter.

Let $K_{n,t}$ be the quantity of capital in country $n$ and period $t$. Then capital dynamics are

$$K_{n,t+1} = (1-\delta_{n,t}) K_{n,t} + I^{\lambda}_{n,t} (\delta_{n,t} K_{n,t})^{1-\lambda}, \tag{6}$$

where $\delta_{n,t}$ is the capital depreciation rate in country $n$ and period $t$ and $\lambda \in [0,1]$ is a parameter governing capital adjustment costs. Solving this for $I_{n,t}$ and viewing it as a function of $K_{n,t}$, $K_{n,t+1}$, and $\delta_{n,t}$, we have

$$I_{n,t} = \Phi(K_{n,t+1}, K_{n,t}; \delta_{n,t}) = \delta^{1-\frac{1}{\lambda}}_{n,t} K_{n,t} \left( \frac{K_{n,t+1}}{K_{n,t}} - (1-\delta_{n,t}) \right)^{\frac{1}{\lambda}}.$$

Take the derivatives of $\Phi$ with respect to the first and the second argument

$$\Phi_1(K_{n,t+1}, K_{n,t}; \delta_{n,t}) = \frac{\partial \Phi}{\partial K_{n,t+1}} (K_{n,t+1}, K_{n,t}; \delta_{n,t}) = \frac{1}{\lambda} \delta^{1-\frac{1}{\lambda}}_{n,t} \left( \frac{K_{n,t+1}}{K_{n,t}} - (1-\delta_{n,t}) \right)^{\frac{1}{\lambda}-1},$$

$$\Phi_2(K_{n,t+1}, K_{n,t}; \delta_{n,t}) = \frac{\partial \Phi}{\partial K_{n,t}} (K_{n,t+1}, K_{n,t}; \delta_{n,t}) = \Phi_1(K_{n,t+1}, K_{n,t}; \delta_{n,t}) \cdot \left( (\lambda-1) \frac{K_{n,t+1}}{K_{n,t}} - \lambda(1-\delta_{n,t}) \right).$$

The dynamic optimization problem of the representative household in country $n$ and period 0 is

$$\max \ (1)$$

subject to (3), (5), and (6). Solving this problem, we obtain the Euler equation

$$\left( \frac{C_{n,t+1}/L_{n,t+1}}{C_{n,t}/L_{n,t}} \right)^{\psi-1} \frac{E_{n,t+1} \bar{\epsilon}_{n,t+1}}{E_{n,t} \bar{\epsilon}_{n,t}} = \beta \frac{\zeta_{n,t+1}}{\zeta_{n,t}} \frac{L_{n,t+1}}{L_{n,t}} \frac{(1-\phi_{n,t+1}) r_{n,t+1} - P^K_{n,t+1} \Phi_2(K_{n,t+2}, K_{n,t+1}; \delta_{n,t+1})}{P^K_{n,t} \Phi_1(K_{n,t+1}, K_{n,t}; \delta_{n,t})}, \tag{7}$$

where

$$\bar{\epsilon}_{n,t} = \sum_{j=a,m,s} \omega^j_{n,t} \epsilon^j, \tag{8}$$

that is, for country $n$ and period $t$, $\bar{\epsilon}_{n,t}$ is the weighted sum of $\{\epsilon_j\}_{j=a,m,s}$ with the expenditure shares $\{\omega^j_{n,t}\}_{j=a,m,s}$ being the weights.

We have described households' behavior thus far. We move on to producers' behavior. The production function of variety $z \in [0,1]$ of sector $j$ in country $n$ and period $t$ is

$$y^j_{n,t}(z) = a^j_{n,t}(z) \left( \frac{K^j_{n,t}(z)}{\gamma^j_{n,t}\alpha^j_{n,t}} \right)^{\gamma^j_{n,t}\alpha^j_{n,t}} \left( \frac{L^j_{n,t}(z)}{\gamma^j_{n,t}(1-\alpha^j_{n,t})} \right)^{\gamma^j_{n,t}(1-\alpha^j_{n,t})} \left( \frac{M^j_{n,t}(z)}{1-\gamma^j_{n,t}} \right)^{1-\gamma^j_{n,t}}. \tag{9}$$

Here $y^j_{n,t}(z)$ is the quantity of output, $a^j_{n,t}(z)$ is the productivity which will be expressed as a realization of a random variable, $K^j_{n,t}(z)$ is the capital, $L^j_{n,t}(z)$ is the labor, $\gamma^j_{n,t} \in (0,1)$ is the value-added share, that is, the cost share on production factors (labor and capital), not on intermediate inputs, $\alpha^j_{n,t} \in (0,1)$ is the cost share on capital *within production factors*, $M^j_{n,t}(z)$ is the CES aggregate of sectoral intermediate goods used for production of variety $z$, that is,

$$M^j_{n,t}(z) = \left( \sum_{j'=a,m,s} (\kappa^{j,j'}_{n,t})^{\frac{1}{\sigma^j}} (M^{j,j'}_{n,t}(z))^{\frac{\sigma^j-1}{\sigma^j}} \right)^{\frac{\sigma^j}{\sigma^j-1}},$$

where $\kappa^{j,j'}_{n,t}$ is the shifter for sector $j$'s demand for sector-$j'$ goods, $M^{j,j'}_{n,t}(z)$ is the input of sector-$j'$ good for production of variety $z$ of sector $j$, and $\sigma^j$ is the elasticity of substitution across sectoral goods for production of sector $j$ goods. In production of sector-$j$ goods, the cost share on sector-$j'$ goods *within intermediate-good costs* is

$$g^{j,j'}_{n,t} = \frac{\kappa^{j,j'}_{n,t}(P^{j'}_{n,t})^{1-\sigma^j}}{\sum_{j"=a,m,s} \kappa^{j,j"}_{n,t}(P^{j"}_{n,t})^{1-\sigma^j}}.$$

The productivity of variety $z$ of sector $j$ in country $n$ and period $t$, $a^j_{n,t}$, follows the Frechét distribution whose the probability distribution function is

$$F^j_{n,t}(a) = \Pr[a^j_{n,t} \le a] = \exp\left[ -\left( \frac{a}{\widetilde{\gamma}^j A^j_{n,t}} \right)^{-\theta^j} \right].$$

Here $\theta^j$ and $A^j_{n,t}$ are the shape parameter and the location parameter of the Frechét distribution, respectively, and $\widetilde{\gamma}^j = [\Gamma((\theta^j + 1 - \eta)/\theta^j)]^{\frac{-1}{1-\eta}}$ with $\Gamma(\cdot)$ being the Gamma function is a normalizing constant. Productivity of varieties are independent within and across sectors, countries, and periods.

Solving the cost minimization problem for the production function (9), the cost for an input bundle is

$$\widetilde{c}^j_{n,t} = (r_{n,t})^{\gamma^j_{n,t}\alpha^j_{n,t}} (w_{n,t})^{\gamma^j_{n,t}(1-\alpha^j_{n,t})} (\xi^j_{n,t})^{1-\gamma^j_{n,t}}, \tag{10}$$

where $\xi^j_{n,t}$ is the CES price index for the composite intermediate good for productio of sector-$j$

goods

$$\xi_{n,t}^{j}=\left[\sum_{j'=a,m,s}\kappa_{n,t}^{j,j'}(P_{n,t}^{j'})^{1-\sigma^{j}}\right]^{\frac{1}{1-\sigma^{j}}}. \tag{11}$$

The price index (or the price of the composite good) of sector $j$ in country $n$ and period $t$ is

$$P_{n,t}^{j}=\left[\sum_{i\in\mathcal{N}}\left(\frac{\tilde{c}_{i,t}^{j}b_{ni,t}^{j}}{A_{i,t}^{j}}\right)^{-\theta^{j}}\right]^{-1/\theta^{j}}, \tag{12}$$

where $b_{ni,t}^{j}$ is the total trade costs including tariffs for goods or services of sector $j$ from country $i$ to country $n$. $b_{ni,t}^{j}$ is expressed as

$$b_{ni,t}^{j}=d_{ni,t}^{j}(1+\tau_{ni,t}^{j}),$$

where $d_{ni,t}^{j}$ is the iceberg trade cost for sector-$j$ goods from country $i$ to country $n$ in period $t$ including phisical trade costs and non-tariff barriers, and $\tau_{ni,t}^{j}$ is country $n$'s tariffs against sector-$j$ goods from country $i$ in period $t$. For later use, define gross tariffs $\tilde{\tau}_{ni,t}^{j}$ by $\tilde{\tau}_{ni,t}^{j}=1+\tau_{ni,t}^{j}$.

The production function of capital (investment) goods in country $n$ and period $t$ is

$$y_{n,t}^{K}=\kappa_{n,t}^{K}\left(\sum_{j=a,m,s}(\kappa_{n,t}^{K,j})^{\frac{1}{\sigma^{K}}}(M_{n,t}^{K,j})^{\frac{\sigma^{K}-1}{\sigma^{K}}}\right)^{\frac{\sigma^{K}}{\sigma^{K}-1}},$$

where $\kappa_{n,t}^{K}$ is the productivity, $M_{n,t}^{K,j}$ is the sector-$j$ goods used for production of capital goods, and $\sigma^{K}$ is the elasticity of substitution across sectoral intermediate goods for production of capital goods. Then the cost share on sector-$j$ goods

$$g_{n,t}^{K,j}=\frac{\kappa_{n,t}^{K,j}(P_{n,t}^{j})^{1-\sigma^{K}}}{\sum_{j'=a,m,s}\kappa_{n,t}^{K,j'}(P_{n,t}^{j'})^{1-\sigma^{K}}}.$$

The ideal price index of capital goods is

$$P_{n,t}^{K}=\frac{1}{\kappa_{n,t}^{K}}\left(\sum_{j=s,m,s}\kappa_{n,t}^{K,j}(P_{n,t}^{j})^{1-\sigma^{K}}\right)^{\frac{1}{1-\sigma^{K}}}.$$

Let $X_{ni,t}^{j}$ be country $n$'s spending on sector $j$ goods (or services) from country $i$ in period $t$. This includes spending for consumption, investment, and intermediate inputs. Summing $X_{ni,t}^{j}$ across $i$, let $X_{n,t}^{j}$ be country $n$'s spending on sector $j$ goods (or services) in period $t$. Let $\pi_{ni,t}^{j}=X_{ni,t}^{j}/X_{n,t}^{j}$, that is, the share of goods from country $i$ within country $n$'s expenditure on sector $j$ goods in period $t$. We call $\pi^j_{ni,t}$ as trade shares following the literature of quantitative trade models. Following Eaton and Kortum (2002), we have

$$\pi^j_{ni,t} = \frac{(\tilde{c}^j_{i,t} b^j_{ni,t}/A^j_{i,t})^{-\theta^j}}{\sum_{i' \in \mathcal{N}} (\tilde{c}^j_{i',t} b^j_{ni',t}/A^j_{i',t})^{-\theta^j}} = \left( \frac{\tilde{c}^j_{i,t} b^j_{ni,t}}{A^j_{i,t} p^j_{n,t}} \right)^{-\theta^j}. \tag{13}$$

Let $Y^j_{n,t}$ be the gross production of sector $j$ in country $n$ and period $t$. It is value not quantity. We have

$$Y^j_{n,t} = \sum_{i \in \mathcal{N}} \frac{\pi^j_{in,t}}{\tilde{\tau}^j_{in,t}} X^j_{i,t}. \tag{14}$$

Country $n$'s spending on sector-$j$ goods in period $t$ consists of the final consumption, the input for production of capital goods, and the input for production of goods and services of various sectors

$$\begin{aligned} X^j_{n,t} &= P^j_{n,t} C^j_{n,t} + g^{K,j}_{n,t} P^K_{n,t} I_{n,t} + \sum_{j'=a,m,s} (1 - \gamma^{j'}_{n,t}) g^{j',j}_{n,t} Y^{j'}_{n,t} \\ &= \omega^j_{n,t} E_{n,t} + + g^{K,j}_{n,t} P^K_{n,t} I_{n,t} + \sum_{j'=a,m,s} (1 - \gamma^{j'}_{n,t}) g^{j',j}_{n,t} Y^{j'}_{n,t}. \end{aligned} \tag{15}$$

In country $n$ and period $t$, the aggregate labor income must be equal to the aggregate labor cost

$$w_{n,t} L_{n,t} = \sum_{j=a,m,s} \gamma^j_{n,t} (1 - \alpha^j_{n,t}) Y^j_{n,t}. \tag{16}$$

Similarly, the aggregate capital income must be equalt to the aggregate capital cost

$$r_{n,t} K_{n,t} = \sum_{j=a,m,s} \gamma^j_{n,t} \alpha^j_{n,t} Y^j_{n,t}. \tag{17}$$

The trade deficit, $D_{n,t}$, is the imports minus the exports

$$D_{n,t} = \underbrace{\sum_{j=a,m,s} \sum_{i=1}^{N} X^j_{n,t} \frac{\pi^j_{ni,t}}{\tilde{\tau}^j_{ni,t}}}_{\text{imports}} - \underbrace{\sum_{j=a,m,s} \sum_{i=1}^{N} X^j_{i,t} \frac{\pi^j_{in,t}}{\tilde{\tau}^j_{in,t}}}_{\text{exports}}. \tag{18}$$

Country $n$'s trade deficit must be equal to its net payment to the global portfolio

$$D_{n,t} = L_{n,t} T^P_t - \phi_{n,t} (w_{n,t} L_{n,t} + r_{n,t} K_{n,t} + \widetilde{T}_{n,t}). \tag{19}$$

Combining (18) and (19), we obtain

$$\sum_{j=a,m,s}\sum_{i=1}^{N}\left(X_{n,t}^{j}\frac{\pi_{ni,t}^{j}}{\widetilde{\tau}_{ni,t}^{j}}-X_{i,t}^{j}\frac{\pi_{in,t}^{j}}{\widetilde{\tau}_{in,t}^{j}}\right)=L_{n,t}T_{t}^{P}-\phi_{n,t}(w_{n,t}L_{n,t}+r_{n,t}K_{n,t}+\widetilde{T}_{n,t}).$$

We move on to the budget balance of the global portfolio. The sum of the net payments from all countries to the global portfolio must be zero

$$\sum_{n=1}^{N}\left\{\phi_{n,t}(w_{n,t}L_{n,t}+r_{n,t}K_{n,t}+\widetilde{T}_{n,t})-L_{n,t}T_{t}^{P}\right\}=0.$$

Solving this for $T_t^P$, we have

$$T_{t}^{P}=\frac{\sum_{n=1}^{N}\phi_{n,t}(w_{n,t}L_{n,t}+r_{n,t}K_{n,t}+\widetilde{T}_{n,t})}{\sum_{n=1}^{N}L_{n,t}}. \tag{20}$$

**Definition 1** (Equilibrium)**.** Given the capital stocks in the initial period $\{K_{n,0}\}_{n\in\mathcal{N}}$, an equilibrium is a tuple of $\{w_{n,t}\}_{n\in\mathcal{N},t=0,\cdots,\infty}$, $\{r_{n,t}\}_{n\in\mathcal{N},t=0,\cdots,\infty}$, $\{E_{n,t}\}_{n\in\mathcal{N},t=0,\cdots,\infty}$, $\{\tilde{c}_{n,t}^{j}\}_{n\in\mathcal{N},t=0,\cdots,\infty,j=a,m,s}$, $\{P_{n,t}^{j}\}_{n\in\mathcal{N},t=0,\cdots,\infty,j=a,m,s}$, $\{\pi_{ni,t}^{j}\}_{(n,i)\in\mathcal{N}\times\mathcal{N},t=0,\cdots,\infty,j=a,m,s}$, $\{Y_{n,t}\}_{n\in\mathcal{N},t=0,\cdots,\infty}$, $\{X_{n,t}^{j}\}_{n\in\mathcal{N},t=0,\cdots,\infty,j=a,m,s}$, $\{\bar{\epsilon}_{n,t}\}_{n\in\mathcal{N},t=0,\cdots,\infty}$, $\{\omega_{n,t}^{j}\}_{n\in\mathcal{N},t=0,\cdots,\infty,j=a,m,s}$, $\{C_{n,t}\}_{n\in\mathcal{N},t=0,\cdots,\infty}$, $\{K_{n,t}\}_{n\in\mathcal{N},t=1,\cdots,\infty}$, $\{I_{n,t}\}_{n\in\mathcal{N},t=0,\cdots,\infty}$, $\{T_{t}^{P}\}_{t=0,\cdots,\infty}$ satisfying a system of equations (3), (4), (5), (6), (7), (8), (10), (12), (13), (14), (15), (16), (17), and (20).

We compute transition paths, that is, equilibria converging to steady states. For this purpose, we define steady states of this model.

**Definition 2** (Steady state)**.** A steady state is an equilibrium in which variables are time-invariant. Specifically, a steady state is a tuple of $\{w_n\}_{n\in\mathcal{N}}$, $\{r_n\}_{n\in\mathcal{N}}$, $\{E_n\}_{n\in\mathcal{N}}$, $\{\tilde{c}_{n}^{j}\}_{n\in\mathcal{N},j=a,m,s}$, $\{P_{n}^{j}\}_{n\in\mathcal{N},j=a,m,s}$, $\{\pi_{ni}^{j}\}_{(n,i)\in\mathcal{N}\times\mathcal{N},j=a,m,s}$, $\{Y_n\}_{n\in\mathcal{N}}$, $\{X_{n}^{j}\}_{n\in\mathcal{N},j=a,m,s}$, $\{\omega_{n}^{j}\}_{n\in\mathcal{N},j=a,m,s}$, $\{C_n\}_{n\in\mathcal{N}}$, $\{K_n\}_{n\in\mathcal{N}}$ satisfying a system of equations (16), (3), (10), (12), (13), (14), (15), (4), (20),

$$r_nK_n=\frac{\alpha}{1-\alpha}w_nL_n,$$

$$r_n=\frac{1-\beta(1-\lambda\delta_n)}{\beta(1-\phi_n)\lambda}P_n^K,$$

and

$$E_n=(1-\phi_n)\left(w_nL_n+r_nK_n+\widetilde{T}_n\right)-\delta_nP_n^KK_n+L_nT^P,$$

dropping time subscripts $t$ from all the equations.

**Steady state.**

# 3 Qualitative Results

Before examining quantitative implications, we qualitatively show how different mechanisms shape the reallocation of economic activities across sectors, in particular, changes in sectoral expenditure share and sectoral value-added share.

## 3.1 Mechanisms for Structural Change

### 3.1.1 Sectoral Expenditure Share

The best consumption measure of structural change is arguably the sectoral expenditure share, $\omega^j_{n,t} = E^j_{n,t}/E_{n,t} = P^j_{n,t}C^j_{n,t}/E_{n,t}$. From consumers' optimal decisions given by (3) and (4), we obtain the following expression for the logarithm changes in $\omega^j_{n,t}$:

$$d\ln\omega^j_{n,t} = \underbrace{(1-\sigma)}_{>0}\Bigg[\underbrace{(1-\omega^j_{n,t})d\ln P^j_{n,t} - \sum_{h\neq j}\omega^h_{n,t}d\ln P^h_{n,t}}_{\text{Baumol effect}} + \underbrace{\left(\epsilon^j - \overline{\epsilon}\right)d\ln\left(\frac{C_{n,t}}{L_{n,t}}\right)}_{\text{Income effect}}\Bigg],$$

where $\overline{\epsilon} \equiv \sum_{h=a,m,s}\omega^h_{n,t}\epsilon^h$ is the budget-share weighted average of $\epsilon^h$ and we assume for simplicity no changes in the demand shifters, $d\ln\Omega^j_{n,t} = 0$. This neatly summarizes the two fundamental mechanisms for structural change at work. To see the Baumol effect captured by the first two terms in the big bracket, suppose the price of sectoral good $j$ falls due to, e.g., increasing sectoral productivity. Given $\sigma$ being less than unity, this leads to a fall in expenditure on the good. The income effect captured by the third term shows that as per-capita consumption $(C_{n,t}/L_{n,t})$ grows, people spend more on good $j$ if the good has a higher (or lower) $\epsilon^j$ than the average $\overline{\epsilon}$, or equivalently an expenditure elasticity higher (or lower) than one.[4] This explains shifts in expenditure from less income-elastic agricultural goods to more income-elastic goods such as manufacturing products and luxurious service. Although not explicit here, international trade plays an important role in determining $\omega^j_{n,t}$ through its effect on goods prices, factor prices, and income.

[4]The expenditure elasticity of good $j$ is given by

$$\frac{\partial\ln C^j_{n,t}}{\partial\ln E_{n,t}} = \eta + (1-\eta)\frac{\epsilon^j}{\overline{\epsilon}},$$

Given $\eta$ being greater than one. the expenditure elasticity is higher (or lower) than one if $\epsilon^j > \overline{\epsilon}$ (or $\epsilon^j < \overline{\epsilon}$).

### 3.1.2 Sectoral Value-Added Share

On the production measure of structural change, any models of structural change have to account for sectoral value-added share defined by

$$\frac{VA^j_{n,t}}{VA_{n,t}} = \frac{r_{n,t}K^j_{n,t} + w_{n,t}L^j_{n,t}}{\sum_{h=a,m,s} r_{n,t}K^h_{n,t} + w_{n,t}L^h_{n,t}} = \frac{\gamma^j_{n,t}Y^j_{n,t}}{\sum_{h=a,m,s}\gamma^h_{n,t}Y^h_{n,t}},$$

for $j = a, m, s$. In our calibration where the capital share in value added is common across sectors, the sectoral value added share coincides with the sectoral share of wage bills.[5] Its logarithm change is given by

$$d\ln\left(\frac{VA^j_{n,t}}{VA_{n,t}}\right) = \left(1 - \frac{VA^j_{n,t}}{VA_{n,t}}\right) d\ln\left(\gamma^j_{n,t}Y^j_{n,t}\right) - \sum_{h\neq j}\frac{VA^h_{n,t}}{VA_{n,t}} d\ln\left(\gamma^h_{n,t}Y^h_{n,t}\right).$$

The share of sector $j$'s value-added naturally increases as capital and labor are more important in producing good $j$, $d\ln\gamma^j_{n,t} > 0$, or gross production of $j$ increases, $d\ln Y^j_{n,t} > 0$, while it decreases as value added in the other sectors grows, $d\ln(\gamma^h_{n,t}Y^h_{n,t}) > 0$ for $h \neq j$.

To see the sectoral gross production $Y^j_{n,t}$ further, we solve the sectoral goods market clearing condition (14) for $Y^j_{n,t}$ to obtain

$$Y^j_{n,t} = \sum_{h=a,m,s}\lambda^{j,h}_{n,t} FD^h_{n,t},$$

where $\lambda^{j,h}_{n,t}$ is a coefficient of the Leontief inverse matrix and shows the value of sector $j$'s gross production induced by a unit increase in sector $h$'s final demand $FD^h_{n,t}$. The sectoral final demand is the sum of sectoral consumption $P^h_{n,t}C^h_{n,t}$, inputs for investment goods $g^{K,h}_{n,t}P^K_{n,t}I_{n,t}$, and net exports $NX^j_{n,t}$ minus tariff revenues $\widetilde{T}^j_{n,t}$, where

$$\begin{aligned}
FD^h_{n,t} &= P^h_{n,t}C^h_{n,t} + g^{K,h}_{n,t}P^K_{n,t}I_{n,t} + NX^h_{n,t} - \widetilde{T}^h_{n,t},\\
NX^h_{n,t} &= \sum_{i=1}^N\left(X^h_{i,t}\frac{\pi^h_{in,t}}{\widetilde{\tau}^h_{in,t}} - X^h_{n,t}\frac{\pi^h_{ni,t}}{\widetilde{\tau}^h_{ni,t}}\right) = -D^h_{n,t},\\
\widetilde{T}^h_{n,t} &= \sum_{i=1}^N X^h_{n,t}\frac{\tau^h_{ni,t}\pi^h_{ni,t}}{\widetilde{\tau}^h_{ni,t}}.
\end{aligned}$$

The logarithmic change of $Y^j_{n,t}$ can be decomposed into changes in the input-output linkage

[5] We see

$$\frac{w_{n,t}L^j_{n,t}}{\sum_{h=a,m,s} w_{n,t}L^h_{n,t}} = \frac{\gamma^j_{n,t}(1-\alpha_{n,t})Y^j_{n,t}}{\sum_{h=a,m,s}\gamma^h_{n,t}(1-\alpha_{n,t})Y^h_{n,t}} = \frac{\gamma^j_{n,t}Y^j_{n,t}}{\sum_{h=a,m,s}\gamma^h_{n,t}Y^h_{n,t}}.$$

and the final demand:

$$d\ln Y_{n,t}^{j} = \underbrace{\sum_{h=a,m,s} y_{n,t}^{j,h} d\ln \lambda_{n,t}^{j,h}}_{\text{Sectoral linkages}} + \underbrace{\sum_{h=a,m,s} y_{n,t}^{j,h} d\ln FD_{n,t}^{h}}_{\text{Final consumption}},$$

$$d\ln FD_{n,t}^{h} = \frac{P_{n,t}^{h}C_{n,t}^{h}}{FD_{n,t}^{h}} d\ln\left(P_{n,t}^{h}C_{n,t}^{h}\right) + \frac{g_{n,t}^{K,h}P_{n,t}^{K}I_{n,t}}{FD_{n,t}^{h}} d\ln\left(g_{n,t}^{K,h}P_{n,t}^{K}I_{n,t}\right) + \frac{NX_{n,t}^{h}}{FD_{n,t}^{h}} d\ln\left(NX_{n,t}^{h}\right) - \frac{\widetilde{T}_{n,t}^{h}}{FD_{n,t}^{h}} d\ln\left(\widetilde{T}_{n,t}^{h}\right),$$

where $y_{n,t}^{j,h} = \lambda_{n,t}^{j,h} FD_{n,t}^{h}/Y_{n,t}^{h}$ is the share of sector $j$'s gross production induced by sector $h$'s final demand. The role of international trade is clear. More net exports in a sector raises the final demand and thus gross production in all sectors through input-output linkages. On the other hand, more imports reduce the final demand and thus gross production as countries rely more on foreign-produced goods.

Among the four determinants of sectoral final demand, preference parameters governing structural change, $\sigma$ and $\epsilon^{j}$, directly affect the change in sectoral consumption:

$$d\ln\left(P_{n,t}^{h}C_{n,t}^{h}\right) = \underbrace{(1-\sigma)}_{>0}\Bigg[\underbrace{\left(1+\frac{\sigma\omega_{n,t}^{h}}{1-\sigma}\right)d\ln P_{n,t}^{j} - \sum_{h\neq j}\frac{\sigma\omega_{n,t}^{h}}{1-\sigma}d\ln P_{n,t}^{h}}_{\text{Baumol effect}} + \underbrace{\left(\epsilon^{j}+\frac{\sigma\bar{\epsilon}_{n,t}}{1-\sigma}\right)d\ln\left(\frac{C_{n,t}}{L_{n,t}}\right)}_{\text{Income effect}}\Bigg] + d\ln L_{n}.$$

The value-added share of sector $j$ decreases if the price of good $j$ falls and the prices of other goods rise. It increases as the per-capita consumption rises, and does so more if the good has a higher $\epsilon^{j}$, or equivalently a higher expenditure elasticity.

Changes in prices and the per-capita consumption are of course shaped by not just own country but also the other countries through international trade. For example, growing trade due to reductions in trade costs allow countries to import goods at lower prices, raising the per-capita consumption in favor of the income effect. On the other hand, growing trade also affects prices in a way that prices in comparative-advantage sectors increase while those in comparative-disadvantage sectors decrease. Therefore, trade counteracts with the Baumol effect by pushing upward the prices of comparative-advantage sectors, i.e., relatively productive sectors.

# 4 Calibration and Solution Algorithm

We bring the model to the data for the global economy. We first describe our main data sources and then discuss the calibration of the structural parameters. We then present the solution algorithm for computing transition paths.

Table 1: List of Countries

| | | | | | |
|---|---|---|---|---|---|
| Australia | Canada | Spain | Greece | Japan | Portugal |
| Austria | China | Finland | India | Korea | Sweden |
| Belgium | Germany | France | Ireland | Mexico | Taiwan |
| Brazil | Denmark | UK | Italy | Netherlands | USA |

### 4.1 Data

Our primary data source is the World Input-Output Database (WIOD) Release 2016 and the Long-Run WIOD (Woltjer et al., 2021; Timmer et al., 2015), which allows us to observe the intermediate input uses across different countries and sectors of both origin and destination. By merging the two datasets, we constructed a database that covers half the century, 1965–2014. Our empirical exercise encompasses 24 countries (see Table 1) and the rest of the world (RoW). They are the listed countries in the Long-Run WIOD, and we moved Hong Kong to the RoW. We aggregate the ISIC industries into three categories as in Table 2. We label D15-D16 Food, Beverages, and Tobacco as agriculture instead of manufacturing due to the nature of its products. Construction and utility supply (e.g., electricity, gas, and water supply) is categorized as a service.[6] We complement the WIOD data with the Penn World Table (PWT) 10.0 (Feenstra et al., 2015) and CEPII Gravity database (Mayer and Zignago, 2011).

### 4.2 Structural Parameters

We begin with discussing our calibration of the parameters in preferences. The discount factor $\beta$ is set at 0.96 to be consistent with a real interest rate of 4 percent per year. We set the inter-temporal elasticity of substitution $\psi = 2$ following Ravikumar et al. (2019). For parameters in the period utility, we choose the elasticities of substitution across sectors $\sigma = 0.5$, and the degree of nonhomotheticity $\epsilon^a = 0.05$ in agriculture, $\epsilon^m = 1$ in manufacturing, and $\epsilon^s = 1.2$ following Comin et al. (2021). $\sigma < 1$ implies that sectoral goods are compliments, and, therefore, the Baumol effect is at work. Values of $\epsilon$s suggest that agriculture is a necessity, service is a luxury, and service starts with a luxury and then becomes a necessity as the consumption expenditure rises.

Value-added share in production function $\gamma^j_{n,t}$ is directly observed in the IO table. Capital share within value-added $\alpha^j_{n,t}$ is calibrated as one minus labor share, which is obtained from the PWT. Since the PWT does not provide the sectoral labor share, we apply the common value across sectors for each year and country. We set the elasticity of substitution across intermediate inputs $\sigma^j = 0.38$ for all $j$ following Atalay (2017). For the capital goods production, we set

[6] Whether the construction and utilities are categorized as manufacturing, service, or an independent sector differs across previous studies. For example, Sposi (2019); Sposi et al. (2021), Uy et al. (2013), Smitkova (2023), Lewis et al. (2022) include construction in the service sector, while Święcki (2017), García-Santana et al. (2021), Herrendorf et al. (2014, 2021), and Betts et al. (2017) include in the manufacturing sector.

Table 2: Three Sectors and Corresponding ISIC3 Codes

| Sector | ISIC3 | Description |
|---|---|---|
| Agriculture | A to B | Agriculture, Hunting, Forestry and Fishing |
| | C | Mining and Quarrying |
| | D15 to 16 | Food, Beverages and Tobacco |
| Manufacturing | D17 to 19 | Textiles, Textile, Leather and Footwear |
| | D21 to 22 | Pulp, Paper, Paper, Printing and Publishing |
| | D23 | Coke, Refined Petroleum and Nuclear Fuel |
| | D24 | Chemicals and Chemical Products |
| | D25 | Rubber and Plastics |
| | D26 | Other Non-Metallic Mineral |
| | D27 to 28 | Basic Metals and Fabricated Metal |
| | D29 | Machinery, Nec |
| | D30 to 33 | Electrical and Optical Equipment |
| | D34 to 35 | Transport Equipment |
| | D n.e.c. | Manufacturing, Nec; Recycling |
| Service | E | Electricity, Gas and Water Supply |
| | F | Construction |
| | G | Wholesale and Retail Trade |
| | H | Hotels and Restaurants |
| | I60 to 63 | Transport and Storage |
| | I64 | Post and Telecommunications |
| | J | Financial Intermediation |
| | K | Real Estate, Renting and Business Activities |
| | L to Q | Community Social and Personal Services |

the elasticity of substitution $\sigma^K = 0.29$ following Sposi et al. (2021). Shape parameters of the Fréchet distribution, i.e., trade elasticities, are chosen as $\theta^a = 8.11$ and $\theta^m = \theta^s = 4.55$. Elasticities for the goods sectors are calibrated based on the estimates of Caliendo and Parro (2015), and we set the elasticity for the service sector to be the same as the manufacturing sector. We will discuss the calibration of productivity and exogenous demand shifters below.

We set the adjustment cost elasticity in the low of motion for capital $\lambda = 0.75$ following Eaton et al. (2016) and the depreciation rate of capital $\delta_{n,t}$ is obtained from the PWT.

### 4.3 Path of Shocks

We back out the trade cost *à la* Head and Ries (2001) assuming the symmetry:

$$d^j_{ni,t} = \left( \frac{\pi^j_{ni,t}\pi^j_{in,t}}{\pi^j_{ii,t}\pi^j_{nn,t}} \right)^{1/(2\theta^j)},$$

where the right-hand-side variables are all observable in the IO table. We calibrate the average productivity $A^j_{n,t}$, i.e., Fréchet location parameters, following Levchenko and Zhang (2016). To begin with, we express the trade share normalized by its own trade share as follows:

$$\frac{\pi^j_{ni,t}}{\pi^j_{nn,t}} = \frac{\left(\frac{\tilde{c}^j_{i,t} d^j_{ni,t}}{A^j_{i,t}}\right)^{-\theta^j}}{\left(\frac{\tilde{c}^j_{n,t}}{A^j_{n,t}}\right)^{-\theta^j}} = \left(\tilde{c}^j_{i,t}/A^j_{i,t}\right)^{-\theta^j} \times \left(\tilde{c}^j_{n,t}/A^j_{n,t}\right)^{\theta^j} \times \left(d^j_{ni,t}\right)^{-\theta^j}.$$

Taking the log of both sides gives:

$$\ln\left(\frac{\pi^j_{ni,t}}{\pi^j_{nn,t}}\right) = \ln\left(\tilde{c}^j_{i,t}/A^j_{i,t}\right)^{-\theta^j} + \ln\left(\tilde{c}^j_{n,t}/A^j_{n,t}\right)^{\theta^j} - \theta^j \ln\left(d^j_{ni,t}\right).$$

We then express the log of the iceberg trade cost using the set of bilateral observables as:

$$\ln\left(d^j_{ni,t}\right) = \mathrm{dist}^j_{k(ni)} + \mathrm{CB}^j_{ni} + \mathrm{CU}^j_{ni,t} + \mathrm{RTA}^j_{ni,t} + ex^j_{it} + \nu^j_{ni,t},$$

where $\mathrm{dist}^j_{k(ni)}$ is the contribution to trade costs of the distance between $n$ and $i$ being in a certain interval[7], $\mathrm{CB}^j_{ni}$ is the indicator if the two countries $n$ and $i$ share the border, $\mathrm{CU}^j_{ni,t}$ indicates if they are in the currency union, $\mathrm{RTA}^j_{ni,t}$ indicates if they are in a regional trade agreement (WTO definition), $ex^j_{it}$ is the exporter fixed effects,[8] and $\nu^j_{ni,t}$ is the bilateral error term. We plug this into the trade share equation (13) and estimate the following using the Pseudo Poisson Maximum Likelihood (PPML) for each sector $j$ while pooling all sampled countries and years:

$$\begin{aligned}\ln\left(\frac{\pi^j_{ni,t}}{\pi^j_{nn,t}}\right) = & \underbrace{\ln\left(\left(\tilde{c}^j_{i,t}/A^j_{i,t}\right)^{-\theta^j} - \theta^j ex^j_{it}\right)}_{\text{exporter-year F.E.}} + \underbrace{\ln\left(\tilde{c}^j_{n,t}/A^j_{i,t}\right)^{\theta^j}}_{\text{importer-year F.F.}} \\ & \underbrace{-\theta^j \ln\left(\mathrm{dist}^j_{k(ni)} + \mathrm{CB}^j_{ni} + \mathrm{CU}^j_{ni,t} + \mathrm{RTA}^j_{ni,t}\right)}_{\text{Bilateral observables}} - \theta^j \nu^j_{ni,t}.\end{aligned}$$

Estimating the gravity equation above allows us to identify the technology-cum-unit-cost term, $\ln\left(\tilde{c}^j_{n,t}/A^j_{i,t}\right)^{\theta^j}$, for each county and year as an importer-year fixed effect, relative to the reference country and year (US in 1965), which we denote by $S^j_{nt} = \left(\tilde{c}^j_{n,t}/A^j_{n,t}\right)^{-\theta^j} / \left(\tilde{c}^j_{US,1965}/A^j_{US,1965}\right)^{-\theta^j}$. Then, we follow Shikher (2013) to recover the sectoral price indices as follows. First, we define

[7]We follow Eaton and Kortum (2002) and intervals are defined, in miles, $[0, 350]$, $[350, 750]$, $[750, 1500]$, $[1500, 3000]$, $[3000, 6000]$, $[6000, \max]$

[8]We follow Waugh (2010).

the own trade share relative to the ones of the reference country and year:

$$\frac{\pi^j_{nn,t}}{\pi^j_{US,US,1965}} = \frac{\left(\tilde{c}^j_{n,t}/A^j_{n,t}\right)^{-\theta^j}}{\left(\tilde{c}^j_{US,1965}/A^j_{US,1965}\right)^{-\theta^j}} \left(\frac{P^j_{n,t}}{P^j_{US,1965}}\right)^{\theta^j}.$$

Since the first term in the right-hand-side is already identified as the importer fixed effect $S^j_{nt}$ in the gravity regression, we can recover the sectoral price indices as follows, for given trade elasticity $\theta^j$:[9]

$$\frac{P^j_{n,t}}{P^j_{US,1965}} = \left(\frac{\pi^j_{nn,t}}{\pi^j_{US,US,1965}} \frac{1}{S^j_{nt}}\right)^{1/\theta^j}.$$

Being armed with the sectoral price indices, we back out the exogenous demand shifter for intermediate inputs, $\kappa^{jh}_{n,t}$, by solving the system of equations for each $j$, $n$, and $t$:

$$g^{j,h}_{n,t} = \frac{\kappa^{j,h}_{n,t}(P^h_{n,t})^{1-\sigma^j}}{\sum_{h'=a,m,s} \kappa^{j,h'}_{n,t}(P^{h'}_{n,t})^{1-\sigma^j}}.$$

by restricting $\sum_{h'} \kappa^{j,h'}_{n,t} = 1$ for each $j$, $n$, and $t$. The left-hand-side of the equation, $g^{j,h}_{n,t}$, is the share of expenditure spent on input from sector $h$ in total input expenditure of $j$, which is directly observed in the IO table. After obtaining $\kappa^{jh}_{n,t}$, we can recover the CES price index for the composite intermediate good $\xi^j_{n,t}$ according to (11). We analogously backout the exogenous demand shifter in the capital goods production function, $\kappa^{Kh}_{n,t}$, by solving the system of equations for each $n$ and $t$:

$$g^{K,h}_{n,t} = \frac{\kappa^{K,h}_{n,t}(P^h_{n,t})^{1-\sigma^K}}{\sum_{h'=a,m,s} \kappa^{K,h'}_{n,t}(P^{h'}_{n,t})^{1-\sigma^K}}.$$

by restricting $\sum_{h'} \kappa^{K,h'}_{n,t} = 1$.

Having $\xi^j_{n,t}$ in hand, we can compute the cost of the input bundle according to (10). To recover wages, we compute the wage bill of each economy by multiplying the labor share obtained from the PWT and the economy-wide value-added computed based on the WIOD. We then divide the wage bill by the total population of the country sourced from the PWT. For the rental price of capital, we divide the return to capital (i.e., total value-added minus the wage bill) by the capital stock. We compute the economy-wide capital stock sequentially over time according to (6), where we obtain the initial period capital stock from the PWT

[9] Note that the price indices are recovered relative to the US in 1965 for each sector. That means the US price index is 1 for all sectors in 1965.

and the investment (gross fixed capital formation) from the WIOD. Then, we can recover the productivity $A^j_{n,t}$ by:[10]

$$A^j_{n,t} = (S^j_{n,t})^{1/\theta^j} \left( \frac{\tilde{c}^j_{n,t}}{\tilde{c}^j_{US,1965}} \right).$$

Using the sectoral price indices computed above, we calibrated the sectoral demand shifter $\Omega^j_{n,t}$ as follows. First, we guess the vector of $\{\Omega^j_{n,t}\}$. Given the data on consumption expenditure $E_{n,t}$ from the WIOD, population $L_{n,t}$ from the PWT, prices $P^j_{n,t}$ and guessed values of $\Omega^j_{n,t}$, solve the consumption index $C^j_{n,t}$ according to (3). Using the computed consumption index, we can find the unique vector of $\Omega^j_{n,t}$ (up to normalization for each $n$ and $t$) by applying the Perron-Frobenius theorem to (4). We then use the value of $\Omega^j_{n,t}$ as the new guess and repeat the steps until we find the fixed points.

Finally, the intertemporal demand shifter $\zeta_{n,t}$ is backed out sequentially according to (7). Using the consumption index $C_{n,t}$ obtained above, we can construct the series of $\zeta_{n,t}$ for each country with the normalizing of initial period $\zeta_{n,1965}$ to be unity.

### 4.4 Values of Shocks

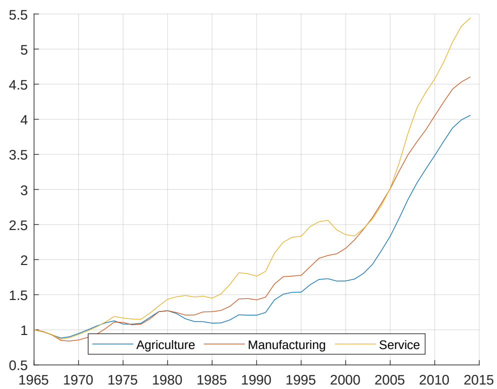
Before moving on to the quantitative results, we summarize the baseline shocks we calibrated above. Figure 2 shows the evolution of sectoral productivity in four countries, China, Germany, Japan, and the US. We normalize the productivity in 1965 to be 1 and take the moving average over 3 years to remove the noise. In Germany, Japan, and the US, the productivity of manufacturing increased more than that of the service sector over the period. For example, manufacturing productivity in the US increased by a factor of 1.8 while the service sector productivity increased by a factor of 1.3. The higher productivity growth in manufacturing than service implies that the expenditure share on manufacturing may drop due to the Baumol effect, even if we do not cosinder impacts of international trade and non-homotheticity preference-driven demand change. In China, service sector productivity grew more, by a factor of 5.5, than manufacturing productivity, by a factor of 4.5. Despite the relatively lower growth in manufacturing productivity to service sector, the manufacturing growth is much higher in level than those in Germany, Japan and the US.

Figure 3 summarizes the evolution of trade costs in the four countries. Here we compute the simple aritmetic average of the bilateral trade costs for each country. Again, we normalize the values in 1965 to be 1 and take the moving average over 3 years. Over the five decades, China and the US observed a sharp drop in service trade costs, almost 70% in China and 90% in the US. However, the major drop in the service trade costs in the two countries happened between 1969 and 1970 of the sample period, and it has been more stable since then. After 1970, service trade costs have dropped only by 20% in China and 25% in the US, which is

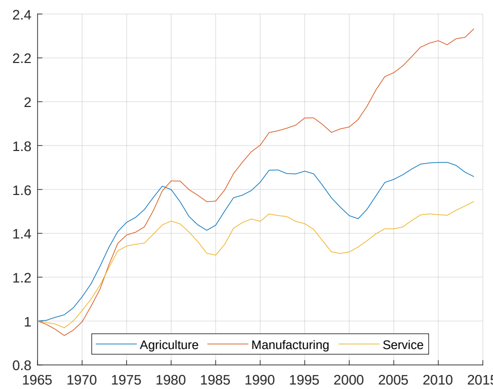
[10] By construction, sectoral productivity takes 1 for the US in 1965 in all sectors.

Figure 2: Productivity Evolution (1965=1, Moving Average)

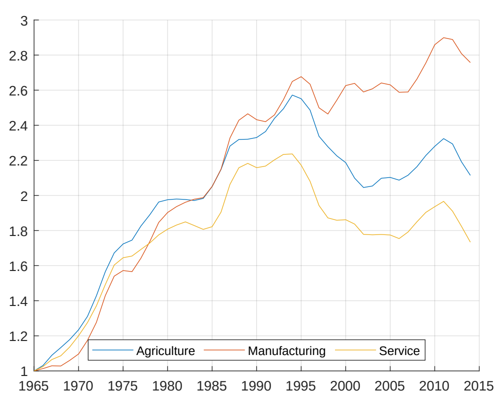
(a) China




(b) Germany




(c) Japan




(d) United States

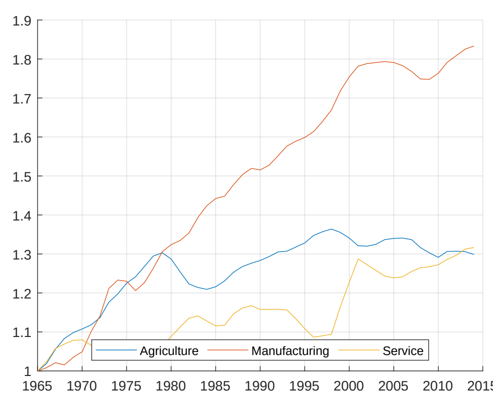
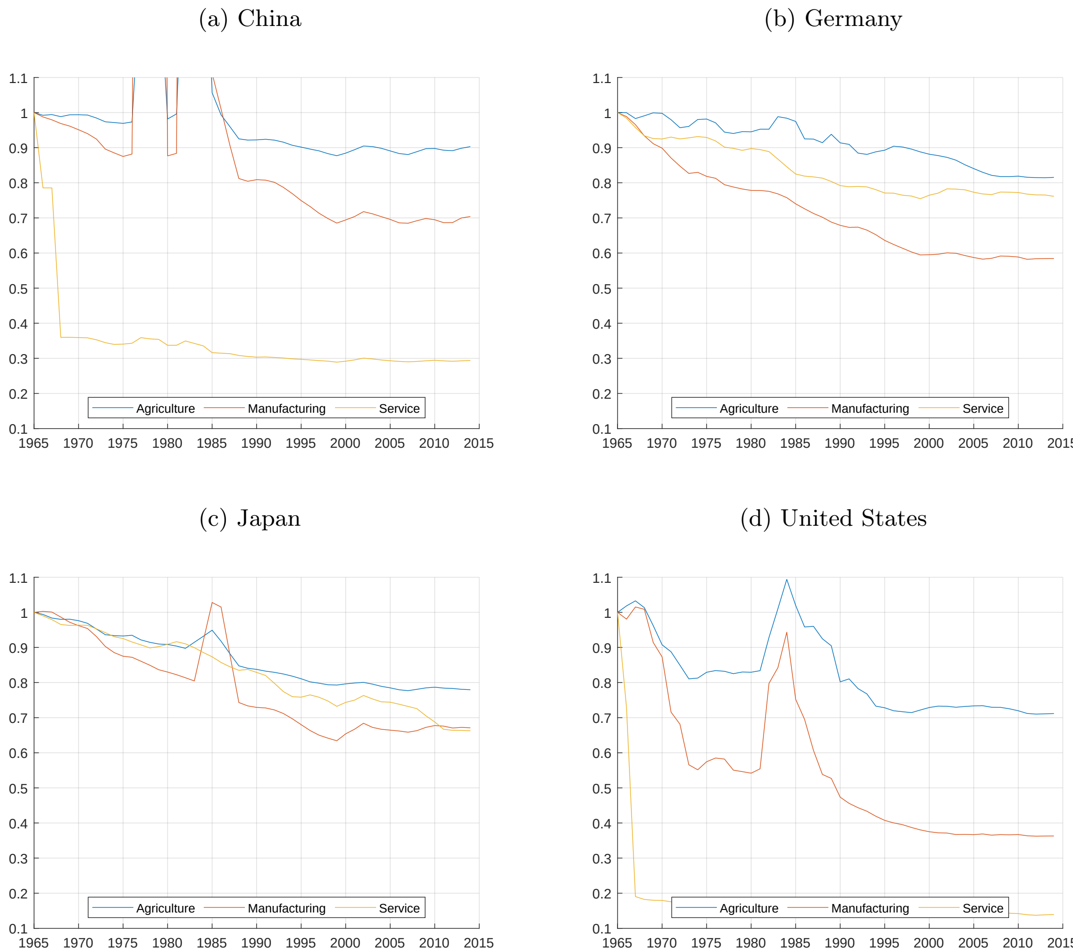


Figure 3: Evolution of Trade Costs (1965=1, Moving Average)

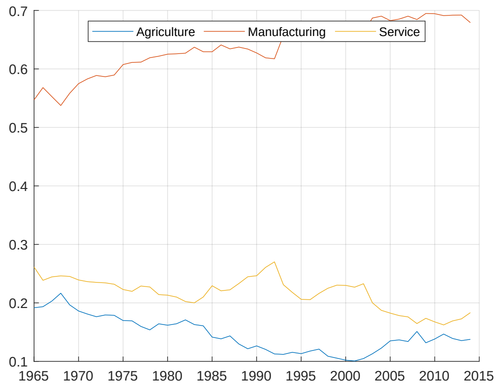



lower than the decline in manufacturing trade costs over the entire sampled period (30% in China and 65% in the US). Germany and Japan also show a large decline in manufacturing trade costs, by 40% and 30%, respectively.

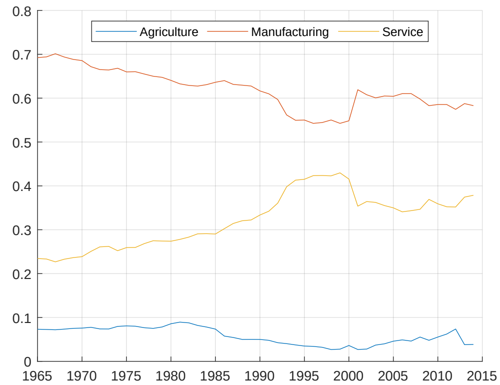
Figure 4 and 5, respectively, demonstrate the intermediate input cost shares for the manufacturing sector and for the capital good production, which are used for the calibration of $\kappa_{n,t}^{h,j}$ and $\kappa_{n,t}^{K,j}$. Except for China, in all three countries, service inputs are becoming more important in manufacturing production. As we saw above, since the productivity of manufacturing grows faster than service sector and the elasticity of substitution across inputs is less than one, the Baumol effect may be a determinant of the increasing share of of service. Growing share of service is also observed for the capital good production in some countries. In addition to the Baumol effect, the nonhomotheticity in the production function may be another reason for the rising share of service. This is beyond the scope of this paper but is an

Figure 4: Cost Shares in Manufacturing Production

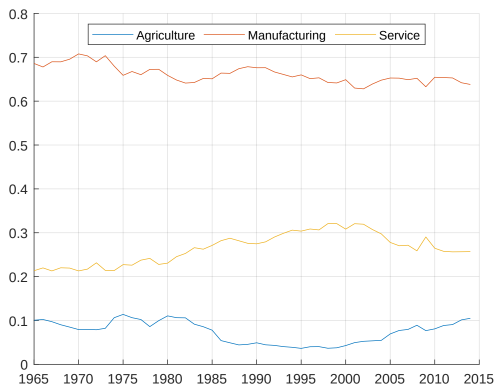
(a) China



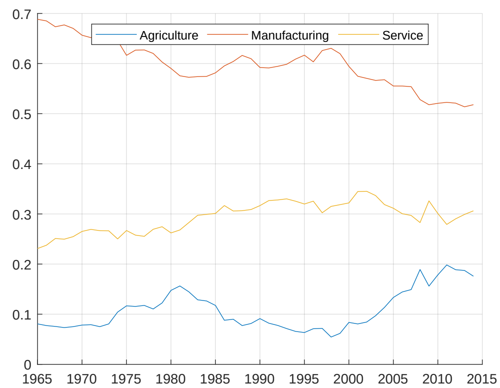

(b) Germany




(c) Japan




(d) United States



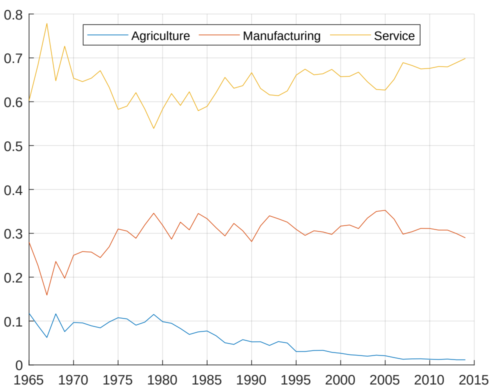

interesting topic for future research.

### 4.5 Solution Algorithm

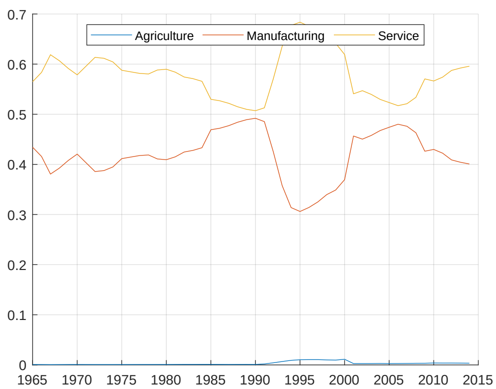
We solve the equilibrium transition path backward. We first solve the model for the steady state according to Definition 2, assuming the 2014 fundamentals (e.g., productivity, trade costs, exogenous demand shifters, etc) last forever. We then suppose that the economy will reach the steady state in 2464 (in 450 years). The solution algorithm for the transition path has two loops: the outer loop finds the sequence of investment (saving) rate $\{\rho_{n,t}\}_{n,t}$ that satisfies the dynamic optimality condition governed by the Euler equation (7) and the inner loop solves the intra-temporal optimization for each period (i.e., solving the sectoral prices and factor prices that satisfy the equilibrium conditions listed in Definition 1). More specifically, For the given sequence of $\{\rho_{n,t}\}_{n,t}$ and the initial period capital stock, we first solve the static equilibrium

Figure 5: Cost Shares in Capital Good Production

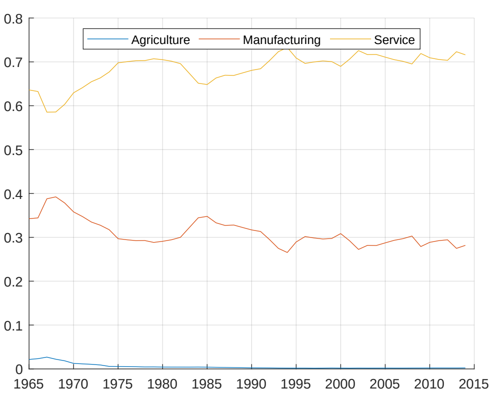
(a) China




(b) Germany




(c) Japan




(d) United States

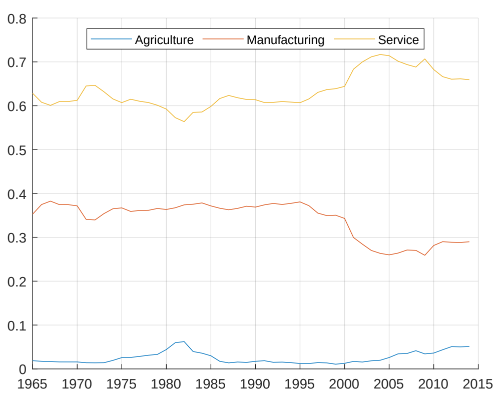
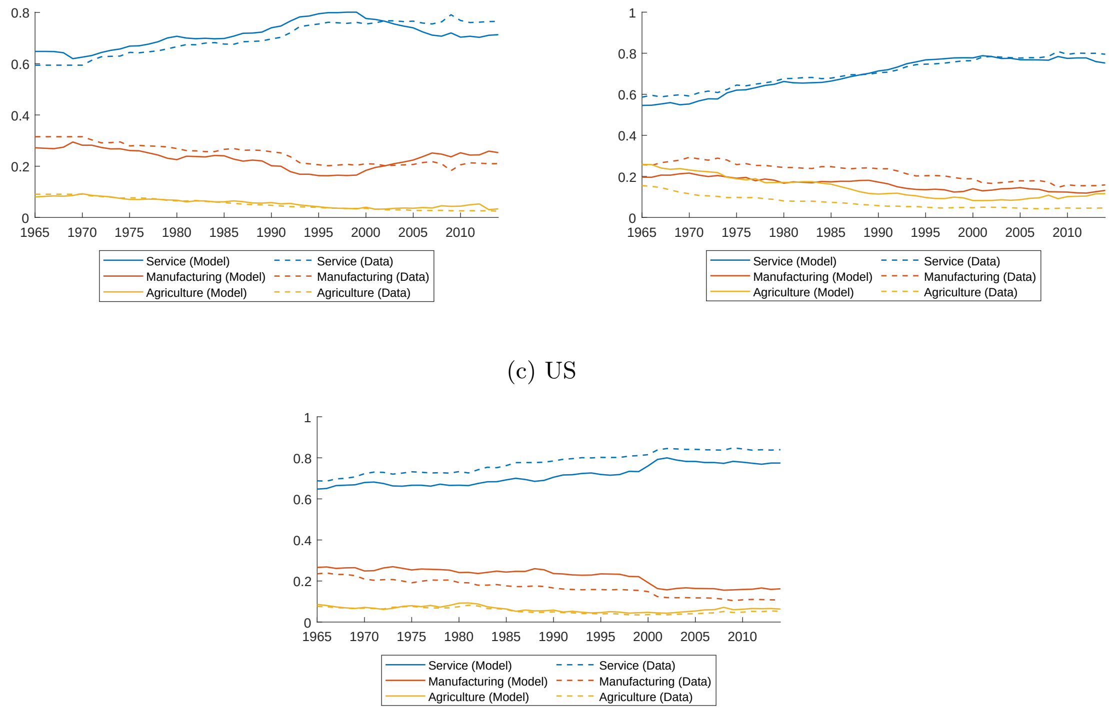


Figure 6: Model Fit: Sectoral Value Added Share in GDP

(a) Germany

(b) Japan

(c) US




period-by-period sequentially from 1965 to 2464. After we solve the periodic equilibrium up to the year 2464, we update $\rho_{n,t}$ backward from 2464 to 1965 according to the Euler equation. Refer to Ravikumar et al. (2019) for the details of the outer loop iteration.

# 5 Quantitative Results

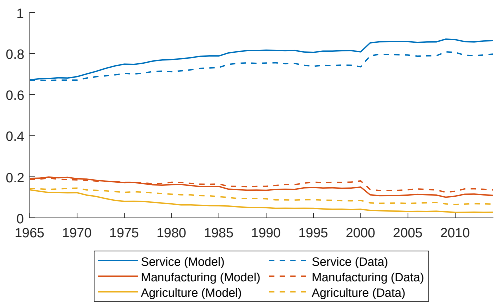
This section presents the quantitative results of the calibrated model.

## 5.1 Fit of the Baseline Model

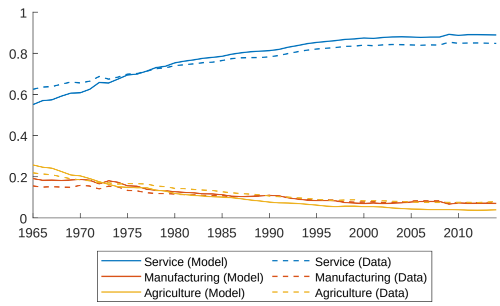
We first show the baseline results. To examine the model's ability to match the data, Figure 6 compares the model-implied (solid lines) value-added share in three sectors (yellow for agriculture, orange for manufacturing, and blue for service) with the data counterpart (dashed lines) for three advanced economies, Germany, Japan, and the US. In the three countries, the model captures the overall trend of falling manufacturing and rising service over time. In Japan, the model over-(under-)predicts the agriculture (manufacturing) value-added share, while in the US, the model over-(under-)predicts the manufacturing (service).

Figure 7: Model Fit: Sectoral Expenditure Share in Final Consumption

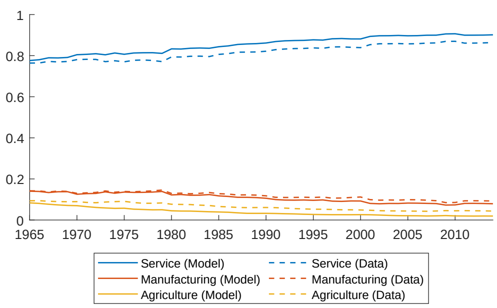
(a) Germany




(b) Japan



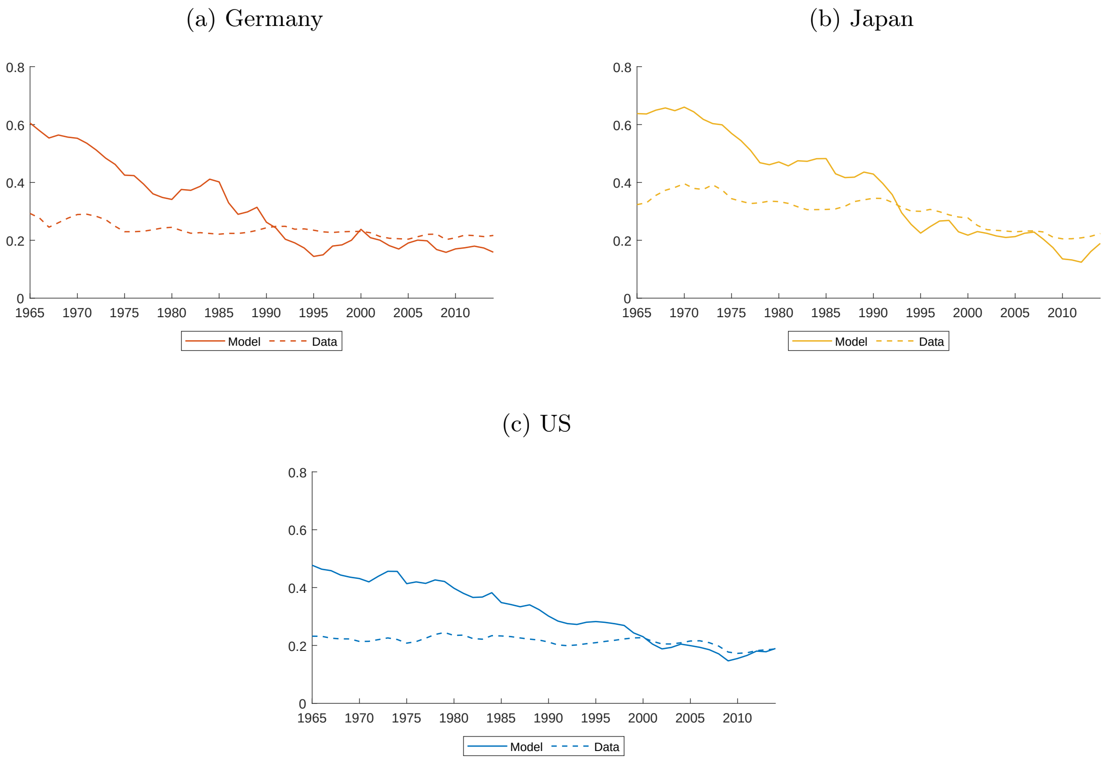

(c) US




Next, we show the expenditure share in final consumption $\omega^j_{n,t}$ implied by the model and the data (Figure 7). In all of the three countries presented, the model accounts for the shift of expenditure from agriculture and manufacturing to service over time. To understand the underlying mechanism of declining manufacturing expenditure share (rising service share), Figure 8 shows the change in sectoral prices over time, where we normalize the prices in the initial period to be one. We see that, in all of the three countries, the manufacturing price drops more than the other two sectors. Due to the complementarity in consumer preferences (i.e., $\sigma < 1$), falling manufacturing prices increase the service expenditure share due to the Baumol effect. Furthermore, nonhomothetic preferences favor rising service expenditures as national income grows over time.

Finally, Figure 8 compares the saving rate in the baseline equilibrium to the data. In all three countries, the model predicts a higher saving rate than the data counterpart in earlier years. The model-implied saving rate gradually falls and converges to the levels close to the data. The result suggests that the initial level of capital stock is off the saddle path to the steady state and, therefore, the economy needs to save more to achieve the higher capital stock.

Figure 8: Model Fit: Saving Rate




## 5.2 Counterfactuals

Now, we will use the model to understand the joint role of international trade and structural change in accounting for the change in sectoral composition among advanced countries. Specifically, we consider the following three counterfactual scenarios.

1. The trade costs between all country pairs are held fixed at the 1965 level forever (1965 trade costs).

2. The productivity in all countries are held fixed at the 1965 level forever (1965 productivity).

3. China's productivity and trade costs with all other countries are held fixed at the 2000 level from 2000 onwards (no China shock).

The first one traces how the changes in trade costs since 1965 contributed to the changes in sectoral composition. The second teases out the contribution of the change in productivity since 1965. The last one illustrates how China's productivity improvement and the decline in trade costs between China and other countries impacted the industry structure in advanced economies.

Figure 9: Transition of Sectoral Prices under Baseline Equilibrium (1965=1)

(a) Germany

(b) Japan

(c) US

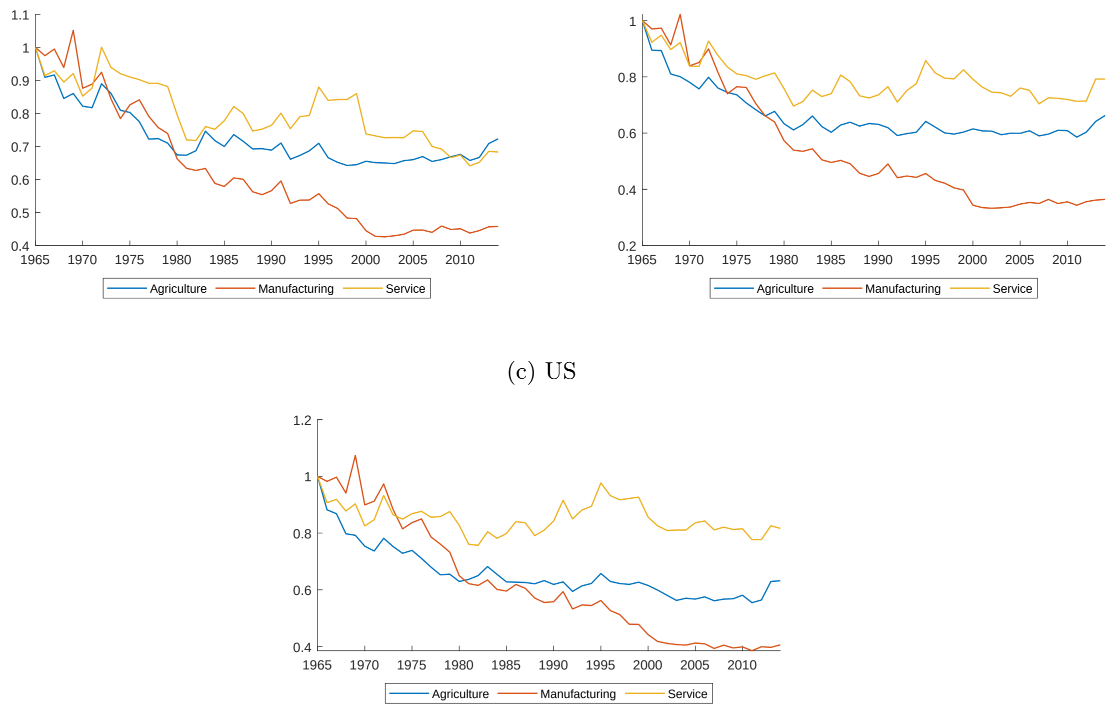



Figure 10 lays out the shares of manufacturing in value-added in the baseline and three counterfactual equilibria for three countries. Solid lines are for the baseline, dashed lines for the first counterfactual with the 1965 trade costs, dotted lines for the second counterfactual with the 1965 productivity, and red lines for the third counterfactual without China shocks.

Before taking a closer look at the results for each country, we will summarize the main findings. First, our results suggest that international trade has heterogeneous impacts on the sectoral composition across countries. Specifically, we find that globalization (i.e., declining trade costs) expanded the manufacturing of the US and Germany while it shrank Japanese manufacturing. Second, in line with the first point, the China shocks have heterogeneous impacts across countries. Most strikingly, in the US, the result suggests that the fall in US manufacturing after 2000 can be entirely attributed to the China shocks.

Let's look at the results for each of the three countries. The top panel, (a), exhibits the manufacturing shares in value-added in Germany. From 1965 to 2000, the baseline and three counterfactuals yield similar value-added shares of manufacturing. Since 2000, however, they diverge. Keeping the productivity or trade costs as in 1965 yields lower manufacturing shares than the baseline. The decline in trade costs since 2000 expanded the relative size of manufacturing in Germany as Germany, the natural manufacture in the global economy

(Eaton and Kortum, 2002), has a comparative advantage in manufacturing. In contrast, the "no China shock" scenario yields higher manufacturing value-added shares than the baseline equilibrium. Therefore, the rise in China's productivity and China's integration into the global economy shrunk the relative size of manufacturing in Germany.[11]

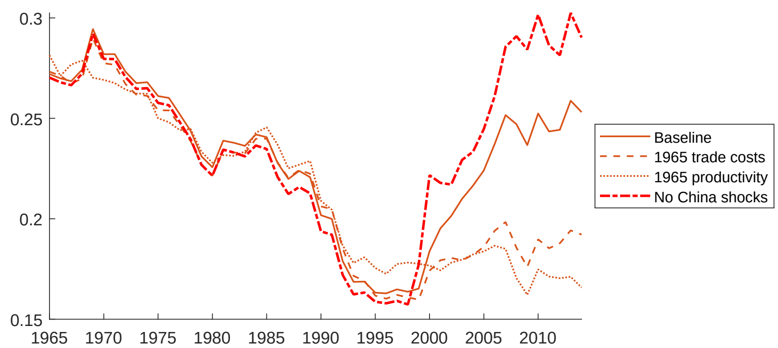
The second panel, (b), plots the manufacturing shares in value-added in Japan. Keeping the trade costs at the 1965 level yields higher manufacturing shares than the baseline. Trade cost reductions would contribute to the growth of manufacturing exports. On the other hand, they would lead to lower-priced imports and push a greater downward pressure on the manufacturing price than the other sectors' prices since manufacturing production relies more on tradable manufacturing inputs. The latter effect dominates the former one so that trade cost reduction from the 1965 level to the current one results in a greater manufacturing share. Keeping the productivity at the 1965 level yields lower manufacturing shares than the baseline. The "no China shock" scenario yields lower manufacturing shares than the baseline until 2014, and they converged. That is, in a medium term, the China shock increased the relative size of manufacturing in Japan. This is consistent with Taniguchi's (2019) empirical finding that import penetration by Chinese products allows the Japanese manufacturers to source lower-priced intermediates and thus to increase employment across Japanese prefectures.

The last panel, (c), shows the manufacturing shares in value-added in the United States. The baseline, 1965 trade costs, and 1965 productivity equilibria all exhibit similar manufacturing shares. However, the gap from the baseline is greater in the equilibrium with the constant 1965 productivity. Productivity growth from the 1965 level to the current one results in a greater manufacturing share. This can be partly explained by the fact that relative productivity in manufacturing to the other sectors is hugely high in 1965 and the subsequent growth narrows the productivity difference. What is striking is the effect of the China shock. The "no China shock" scenario yields largely constant manufacturing shares. Therefore, virtually all of the decline in the manufacturing share since 2000 is attributable to the China shocks.

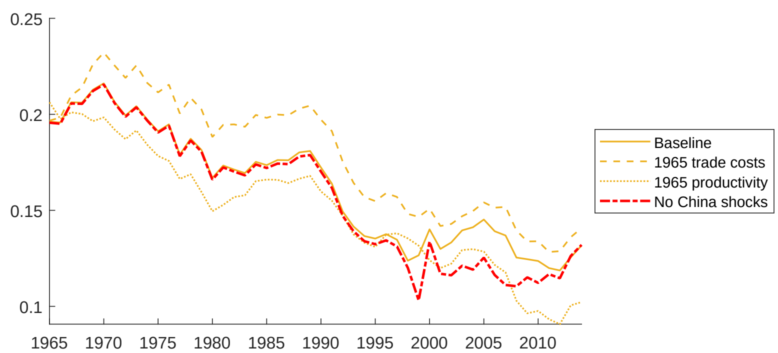
[11] On the other hand, Dauth et al. (2014) found that China's trade integration retained the manufacturing in Germany. The difference between their result and ours may be partly due to the general equilibrium effect of the world trade market.

Figure 10: Counterfactual Manufacturing Shares in Value-Added

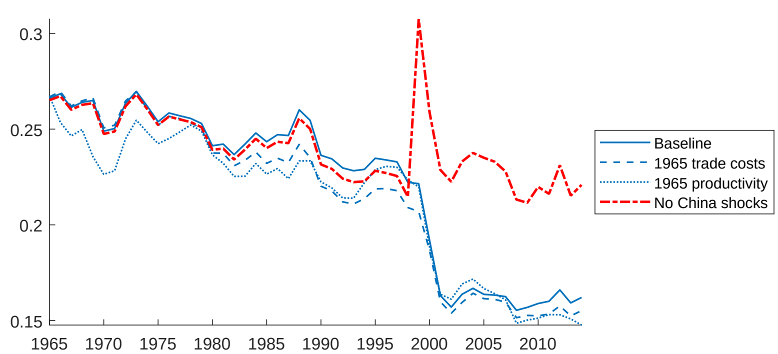
(a) Germany




(b) Japan




(c) US




# 6 Conclusion

We developed a dynamic general equilibrium model that features international trade, capital accumulation, sector-biased productivity growth, and non-homothetic preferences to dissect the evolution of sectoral composition in the global economy. We bring the model to the data for the world economy with 24 countries over the period of 1965–2014. Our calibrated model captures the declining share of manufacturing and rising share of service in value-added in the US. We undertook a few counterfactual experiments to explore the role of the evolution of productivity and trade costs since 1965 and the China shock. In Germany and the US, the China shock reduced the manufacturing share in value-added. Especially in the US, the China shock fully explained the decline in the manufacturing share since 2000. In contrast, in Japan, the China shock increased the manufacturing share.

A few more next steps are in order. First of all, due to the increasing saving rate implied by the model, the model fails to match the evolution of expenditure shares across sectors. We plan to fix the too-high saving rate by introducing a time-country specific demand shifter as in Eaton et al. (2016); Sposi et al. (2021) to better fit the model to the data. Then, we will provide a more detailed decomposition of the effect of falling trade costs on sectoral composition according to the analytical formulas developed in Section 3. Our expected results answer questions such as: to what extent globalization would reduce manufacturing if none of the forces of structural change were in operation (e.g., non-homotheticity of consumer preferences); whether the decline in manufacturing resulting from globalization has similar welfare implications to the one resulting from structural change. Furthermore, we plan to apply our framework to study a few globalization episodes mentioned in the Introduction, in particular, the impact of the eastward enlargement of the EU in 2004 and 2007 on sectoral composition and trade patterns among the EU member states.

## References


Acemoglu, D. (2008). *Introduction to Modern Economic Growth*. Princeton University Press, Princeton.

Acemoglu, D., Autor, D., Dorn, D., Hanson, G. H., and Price, B. (2016). Import competition and the great US employment sag of the 2000s. *Journal of Labor Economics*, 34(S1):S141–S198.

Albouy, D., Chernoff, A., Lutz, C., and Warman, C. (2019). Local labor markets in Canada and the United States. *Journal of Labor Economics*, 37(S2):S533–S594.

Alessandria, G., Johnson, R. C., and Yi, K.-M. (2023). Perspectives on trade and structural transformation. *Oxford Development Studies*, pages 1–21.

Atalay, E. (2017). How important are sectoral shocks? *American Economic Journal: Macroeconomics*, 9(4):254–80.

Autor, D. H., Dorn, D., and Hanson, G. H. (2013). The China syndrome: Local labor market effects of import competition in the United States. *American Economic Review*, 103(6):2121–68.

Autor, D. H., Dorn, D., and Hanson, G. H. (2016). The China shock: Learning from labor-market adjustment to large changes in trade. *Annual Review of Economics*, 8:205–240.

Baumol, W. J. (1967). Macroeconomics of unbalanced growth: The anatomy of urban crisis. *The American economic review*, 57(3):415–426.

Betts, C., Giri, R., and Verma, R. (2017). Trade, reform, and structural transformation in south korea. *IMF Economic Review*, 65:745–791.

Caliendo, L., Dvorkin, M., and Parro, F. (2019). Trade and labor market dynamics: General equilibrium analysis of the China trade shock. *Econometrica*, 87(3):741–835.

Caliendo, L. and Parro, F. (2015). Estimates of the trade and welfare effects of NAFTA. *The Review of Economic Studies*, 82(1):1–44.

Comin, D., Lashkari, D., and Mestieri, M. (2021). Structural change with long-run income and price effects. *Econometrica*, 89(1):311–374.

Dauth, W., Findeisen, S., and Suedekum, J. (2014). The rise of the East and the Far East: German labor markets and trade integration. *Journal of the European Economic Association*, 12(6):1643–1675.

Dekle, R., Eaton, J., and Kortum, S. (2008). Global rebalancing with gravity: Measuring the burden of adjustment. *IMF Staff Papers*, 55(3):511–540.

Eaton, J. and Kortum, S. (2002). Technology, geography, and trade. *Econometrica*, 70(5):1741–1779.

Eaton, J., Kortum, S., Neiman, B., and Romalis, J. (2016). Trade and the global recession. *American Economic Review*, 106(11):3401–38.

Feenstra, R. C., Inklaar, R., and Timmer, M. P. (2015). The next generation of the Penn World Table. *American Economic Review*, 105(10):3150–82.

García-Santana, M., Pijoan-Mas, J., and Villacorta, L. (2021). Investment demand and structural change. *Econometrica*, 89(6):2751–2785.

Hanoch, G. (1975). Production and demand models with direct or indirect implicit additivity. *Econometrica*, 43(3):395–419.

Head, K. and Ries, J. (2001). Increasing returns versus national product differentiation as an explanation for the pattern of U.S.-Canada trade. *American Economic Review*, 91(4):858–876.

Herrendorf, B., Rogerson, R., and Valentinyi, A. (2014). Growth and structural transformation. *Handbook of Economic Growth*, 2:855–941.

Herrendorf, B., Rogerson, R., and Valentinyi, A. (2021). Structural change in investment and consumption—A unified analysis. *The Review of Economic Studies*, 88(3):1311–1346.

Kehoe, T. J., Ruhl, K. J., and Steinberg, J. B. (2018). Global imbalances and structural change in the United States. *Journal of Political Economy*, 126(2):761–796.

Kongsamut, P., Rebelo, S., and Xie, D. (2001). Beyond balanced growth. *The Review of Economic Studies*, 68(4):869–882.

Kuznets, S. (1973). Modern economic growth: Findings and reflections. *The American Economic Review*, 63(3):247–258.

Levchenko, A. A. and Zhang, J. (2016). The evolution of comparative advantage: Measurement and welfare implications. *Journal of Monetary Economics*, 78(C):96–111.

Lewis, L. T., Monarch, R., Sposi, M., and Zhang, J. (2022). Structural change and global trade. *Journal of the European Economic Association*, 20(1):476–512.

Malgouyres, C. (2017). The impact of Chinese import competition on the local structure of employment and wages: Evidence from France. *Journal of Regional Science*, 57(3):411–441.

Matsuyama, K. (2019). Engel's law in the global economy: Demand-induced patterns of structural change, innovation, and trade. *Econometrica*, 87(2):497–528.

Mayer, T. and Zignago, S. (2011). Notes on CEPII's distances measures: The geodist database. Working Papers 2011-25, CEPII.

Ngai, L. R. and Pissarides, C. A. (2007). Structural change in a multisector model of growth. *American Economic Review*, 97(1):429–443.

Pierce, J. R. and Schott, P. K. (2016). The surprisingly swift decline of US manufacturing employment. *American Economic Review*, 106(7):1632–1662.

Ravikumar, B., Santacreu, A. M., and Sposi, M. (2019). Capital accumulation and dynamic gains from trade. *Journal of International Economics*, 119(C):93–110.

Rodrik, D. (2016). Premature deindustrialization. *Journal of Economic Growth*, 21:1–33.

Shikher, S. (2013). Determinants of trade and specialization in the organisation for economic co-operation and development countries. *Economic Inquiry*, 51(1):138–158.

Smitkova, L. (2023). Structural change in an open economy. mimemo.

Sposi, M. (2019). Evolving comparative advantage, sectoral linkages, and structural change. *Journal of Monetary Economics*, 103:75–87.

Sposi, M., Yi, K.-M., and Zhang, J. (2021). Deindustrialization and industry polarization. Working Paper 29483, National Bureau of Economic Research.

Święcki, T. (2017). Determinants of structural change. *Review of Economic Dynamics*, 24:95–131.

Taniguchi, M. (2019). The effect of an increase in imports from China on local labor markets in Japan. *Journal of the Japanese and International Economies*, 51:1–18.

Timmer, M. P., Dietzenbacher, E., Los, B., Stehrer, R., and de Vries, G. J. (2015). An illustrated user guide to the world input–output database: the case of global automotive production. *Review of International Economics*, 23(3):575–605.

Utar, H. (2018). Workers beneath the floodgates: Low-wage import competition and workers' adjustment. *Review of Economics and Statistics*, 100(4):631–647.

Uy, T., Yi, K.-M., and Zhang, J. (2013). Structural change in an open economy. *Journal of Monetary Economics*, 60(6):667–682.

Waugh, M. E. (2010). International trade and income differences. *American Economic Review*, 100(5):2093–2124.

Woltjer, P., Gouma, R., and Timmer, M. P. (2021). Long-run world input-output database: Version 1.1 sources and methods. *GGDC Research Memorandum*, (190):1–56.